

Fashion_minst training and validation.

From your Linux terminal

- Activate your own conda environment
- import all packages needed to perform deep learning with Keras
- Load database Fashion_Mnist

The dataset is already divided into a practice set and a test set.

- Create a validation set.
- Scale the input characteristics, in order to drive the network by the gradient descent algorithm.
- Scale the pixel intensities so that they belong into [0,1].
- Name the 10 classes in the dataset
- Build a neural network with 2 hidden layers with the sequential API'
 - Layer 1 with 300 neurons. Each neuron admitting the ReLu function as activation function
 - Layer 2 with 100 neurons. Each neuron admitting the ReLu function as activation function
 - Output layer with 10 neurons and Softmax as activation function'
- Summarize the model
- Compile the model

Training : Train the model with the fit() method. Use Sparse_categorical_crossentropy as a loss function and the stochastic gradient descent (SGD) as optimizer.

- Use the plot() method after importing Panda, to plot the learning curve.
- Evaluate the model
- Use your as a classifier on images that have not been used for training.
- Test other network settings with other hidden layers, etc.